

DSA Roadmap for a 6-Year Java Developer

Arrays & Strings (25)

1. Two Sum
2. Best Time to Buy and Sell Stock
3. Contains Duplicate
4. Product of Array Except Self
5. Maximum Subarray
6. Merge Sorted Array
7. Remove Duplicates from Sorted Array
8. Move Zeroes
9. Rotate Array
10. Majority Element
11. Valid Anagram
12. Group Anagrams
13. Longest Common Prefix
14. Valid Palindrome
15. Reverse String
16. Reverse Words in a String
17. String Compression
18. Longest Substring Without Repeating Characters
19. Longest Palindromic Substring
20. Encode and Decode Strings
21. 3Sum
22. Container With Most Water
23. Trapping Rain Water
24. Minimum Window Substring
25. Find All Anagrams in a String

Binary Search (18)

1. Binary Search
2. Search Insert Position
3. Guess Number Higher or Lower
4. First Bad Version
5. Search in Rotated Sorted Array
6. Find Minimum in Rotated Sorted Array
7. Peak Index in Mountain Array
8. Find Peak Element
9. Search a 2D Matrix
10. Koko Eating Bananas
11. Capacity To Ship Packages Within D Days
12. Split Array Largest Sum
13. Median of Two Sorted Arrays
14. Time Based Key Value Store
15. Find First and Last Position
16. Single Element in Sorted Array
17. Sqrt(x)
18. Aggressive Cows

HashMap/HashSet (15)

1. Two Sum
2. Contains Duplicate
3. Valid Anagram
4. Intersection of Two Arrays
5. Happy Number
6. Isomorphic Strings
7. Word Pattern
8. Longest Consecutive Sequence
9. Top K Frequent Elements
10. Subarray Sum Equals K
11. Ransom Note
12. Jewels and Stones
13. Find Duplicate Number
14. First Unique Character
15. Roman to Integer

Two Pointers (15)

1. Valid Palindrome
2. Two Sum II
3. Reverse String
4. Move Zeroes
5. Remove Duplicates
6. Container With Most Water
7. 3Sum
8. 4Sum
9. Sort Colors
10. Squares of Sorted Array
11. Backspace String Compare
12. Merge Strings Alternately
13. Partition Labels
14. Boats to Save People
15. Minimum Difference Between Highest and Lowest

Sliding Window (15)

1. Maximum Average Subarray
2. Longest Substring Without Repeating Characters
3. Permutation in String
4. Find All Anagrams
5. Minimum Window Substring
6. Longest Repeating Character Replacement
7. Max Consecutive Ones III
8. Fruit Into Baskets
9. Subarray Product Less Than K
10. Minimum Size Subarray Sum
11. Sliding Window Maximum
12. Grumpy Bookstore Owner
13. Binary Subarrays With Sum
14. Number of Nice Subarrays
15. Longest Ones

Linked List (15)

1. Reverse Linked List
2. Middle of Linked List
3. Linked List Cycle
4. Palindrome Linked List
5. Merge Two Sorted Lists
6. Remove Nth Node From End
7. Intersection of Two Linked Lists
8. Reorder List
9. Copy List with Random Pointer
10. Add Two Numbers
11. Swap Nodes in Pairs
12. Rotate List
13. Sort List
14. LRU Cache
15. Flatten Multilevel Doubly Linked List

Stack & Queue (15)

1. Valid Parentheses
2. Min Stack
3. Implement Queue Using Stacks
4. Implement Stack Using Queues
5. Daily Temperatures
6. Next Greater Element
7. Evaluate Reverse Polish Notation
8. Baseball Game
9. Decode String
10. Simplify Path
11. Asteroid Collision
12. Largest Rectangle in Histogram
13. Sliding Window Maximum
14. Generate Parentheses
15. Car Fleet

Trees & BST (28)

1. Maximum Depth
2. Same Tree
3. Invert Binary Tree
4. Balanced Binary Tree
5. Diameter of Binary Tree
6. Path Sum
7. Binary Tree Level Order Traversal
8. Zigzag Traversal
9. Right Side View
10. Lowest Common Ancestor BST
11. Validate BST
12. Kth Smallest Element
13. BST Iterator
14. Construct Tree from Traversals
15. Serialize Deserialize Tree
16. Subtree of Another Tree
17. Symmetric Tree
18. Count Good Nodes
19. Maximum Path Sum
20. House Robber III
21. Binary Tree Paths
22. Flatten Binary Tree
23. Boundary Traversal
24. Vertical Order Traversal
25. Top View
26. Bottom View
27. Morris Traversal
28. Recover BST

Heap/Priority Queue (15)

1. Kth Largest Element
2. Top K Frequent Elements
3. Find Median From Data Stream
4. Merge K Sorted Lists
5. Last Stone Weight
6. K Closest Points
7. Task Scheduler
8. Reorganize String
9. Meeting Rooms II
10. Smallest Range
11. IPO
12. Connect Ropes
13. Sort Characters By Frequency
14. Ugly Number II
15. Kth Smallest in Matrix

Recursion & Backtracking (22)

1. Factorial
2. Fibonacci
3. Power Function
4. Generate Parentheses
5. Subsets
6. Subsets II
7. Permutations
8. Permutations II
9. Combination Sum
10. Combination Sum II
11. Letter Combinations of Phone Number
12. Palindrome Partitioning
13. N Queens
14. Word Search
15. Sudoku Solver
16. Restore IP Addresses
17. Rat in a Maze
18. Knight Tour
19. Tower of Hanoi
20. M Coloring Problem
21. Combination Sum III
22. Beautiful Arrangement

Graphs (28)

1. Number of Islands
2. Clone Graph
3. Flood Fill
4. Course Schedule
5. Course Schedule II
6. Rotting Oranges
7. Pacific Atlantic Water Flow
8. Graph Valid Tree
9. Redundant Connection
10. Network Delay Time
11. Dijkstra Shortest Path
12. Bellman Ford
13. Topological Sort
14. Word Ladder
15. Alien Dictionary
16. Surrounded Regions
17. Accounts Merge
18. Min Cost to Connect Points
19. Cheapest Flights Within K Stops
20. BFS Traversal
21. DFS Traversal
22. Detect Cycle Undirected
23. Detect Cycle Directed
24. Kosaraju Algorithm
25. Tarjan Bridges
26. Strongly Connected Components
27. Disjoint Set Union
28. Number of Provinces

Dynamic Programming (28)

1. Climbing Stairs
2. House Robber
3. House Robber II
4. Coin Change
5. Coin Change II
6. Longest Increasing Subsequence
7. Longest Common Subsequence
8. Edit Distance
9. 0/1 Knapsack
10. Unbounded Knapsack
11. Partition Equal Subset Sum
12. Target Sum
13. Decode Ways
14. Unique Paths
15. Minimum Path Sum
16. Triangle
17. Maximum Product Subarray
18. Word Break
19. Palindromic Substrings
20. Longest Palindromic Subsequence
21. Burst Balloons
22. Matrix Chain Multiplication
23. Rod Cutting
24. Egg Dropping
25. Best Time to Buy and Sell Stock with Cooldown
26. Distinct Subsequences
27. Interleaving String
28. Regular Expression Matching